DO NOW — RECALL

Lesson 079 · Bubble Sort: Adjacent Swaps and Passes

Name: _____ Class: _____ Date: _____

1. Which elements does bubble sort compare?

2. What is placed by one ascending pass?

3. What does swap do?

4. Swap elements at indexes 0 and 1 in array [4,2,6]. Give the new array.

DO NOW — CHECK / OBJECTIVES

Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

1. Model answer

Adjacent pairs.

2. Model answer

Largest unsorted value at end.

3. Model answer

Exchanges two stored values.

4. Model answer

[2,4,6].

By the end, you can…

• Compare adjacent pairs, swapping them when out of ascending order.

• One completed ascending pass places the largest remaining element at the right end of the unsorted prefix.

• Straightforward bubble sort is $O(n^2)$ worst time and O(1) extra memory; an early-exit flag can give O(n) best time.

Vocabulary — know these terms

bubble sort — Compare/swap adjacent values.

pass — Sweep through neighbouring pairs.

swap — Exchange two elements.

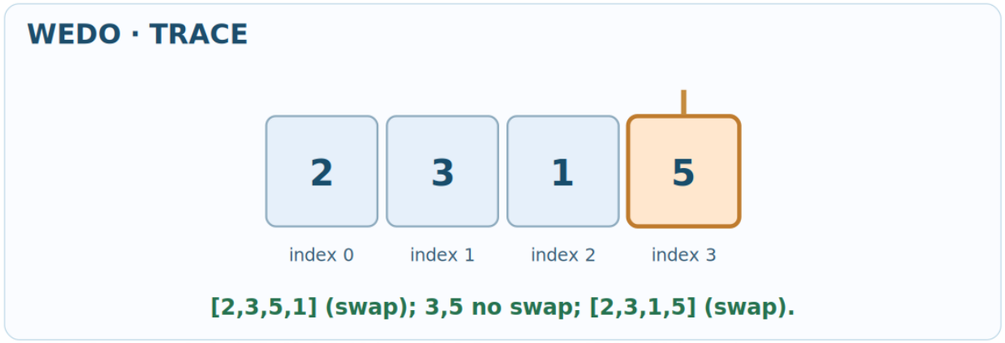
invariant — Always true state after given stage.

We Do source CHECK · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

Show full first pass [3,2,5,1].

[2,3,5,1] (swap); 3,5 no swap; [2,3,1,5] (swap).



WEDO · TRACE; array 2, 3, 1, 5; [2,3,5,1] (swap); 3,5 no swap; [2,3,1,5] (swap).

NEW LEARNING

Read, annotate and explain

Understand

Compare adjacent pairs, swapping them when out of ascending order.

Trace

One completed ascending pass places the largest remaining element at the right end of the unsorted prefix.

Explain

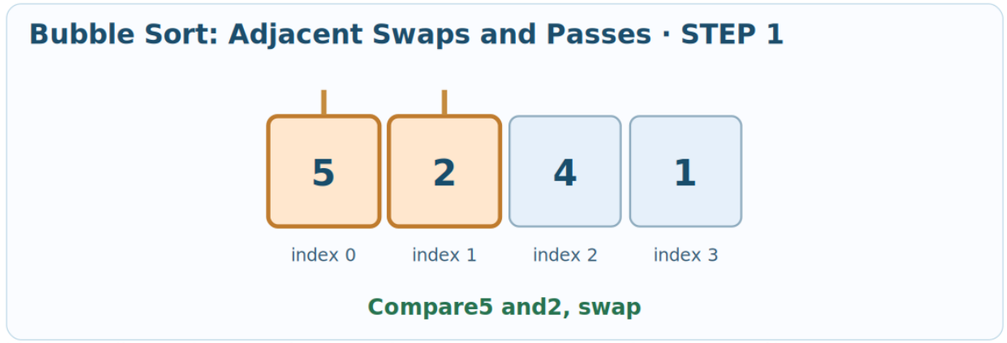
Straightforward bubble sort is $O(n^2)$ worst time and O(1) extra memory; an early-exit flag can give O(n) best time.

Explain one step in your own words, using the correct technical terms.

New Learning · original lesson diagrams

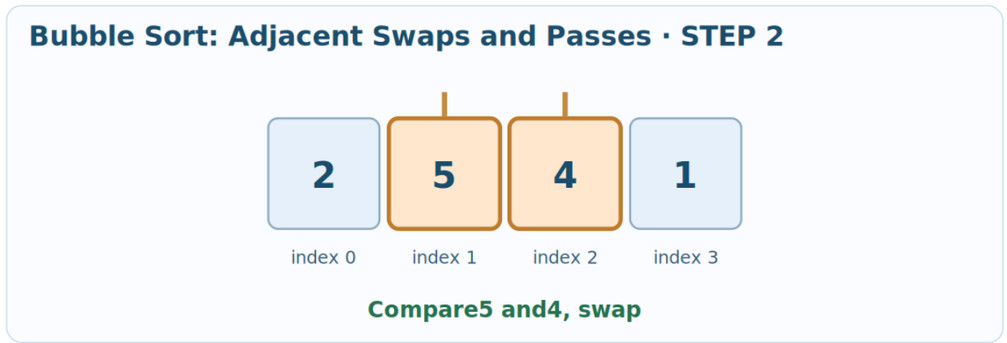
Diagrams appear after the related explanation. Annotate the original source visual.

New Learning



Bubble Sort: Adjacent Swaps and Passes · STEP 1; array 5, 2, 4, 1; Compare5 and2, swap

New Learning



Bubble Sort: Adjacent Swaps and Passes · STEP 2; array 2, 5, 4, 1; Compare5 and4, swap

WE DO — CHECK AND IMPROVE

Source-grounded model answers

CHECK only after completing the guided practice and Exit Ticket. Use GREEN PEN to improve your method and explain any error.

Guided 1 — CHECK

Check the working
[2,3,5,1] (swap); 3,5 no swap; [2,3,1,5] (swap).

EXIT TICKET / REFLECTION

Check your understanding

Complete the actual lesson Exit Ticket before checking.

After ascending first bubble pass, smallest value must be fixed at index0.

Circle: Confident / Getting there / Need help

Explain what you can now do independently.

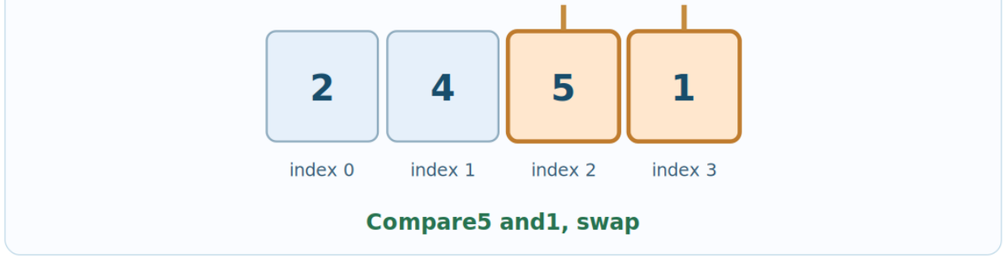
GREEN PEN — Correct and explain

Choose one error. Write the correct method and explain precisely why your first response was wrong.

New Learning · original lesson diagrams

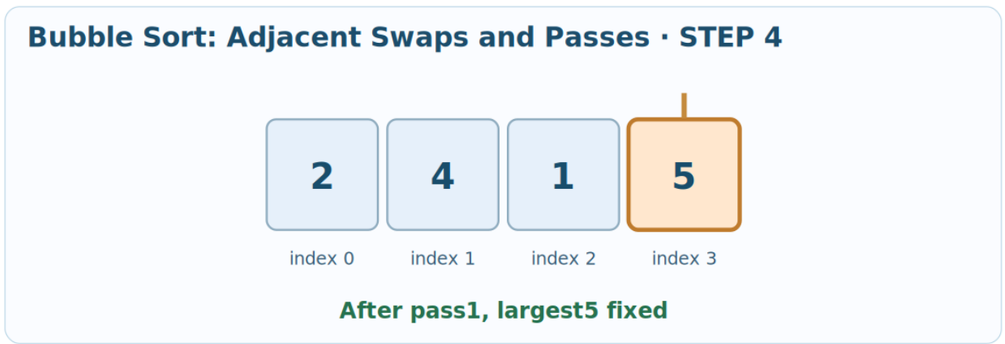
Diagrams appear after the related explanation. Annotate the original source visual.

New Learning



Bubble Sort: Adjacent Swaps and Passes · STEP 3; array 2, 4, 5, 1; Compare5 and1, swap

New Learning

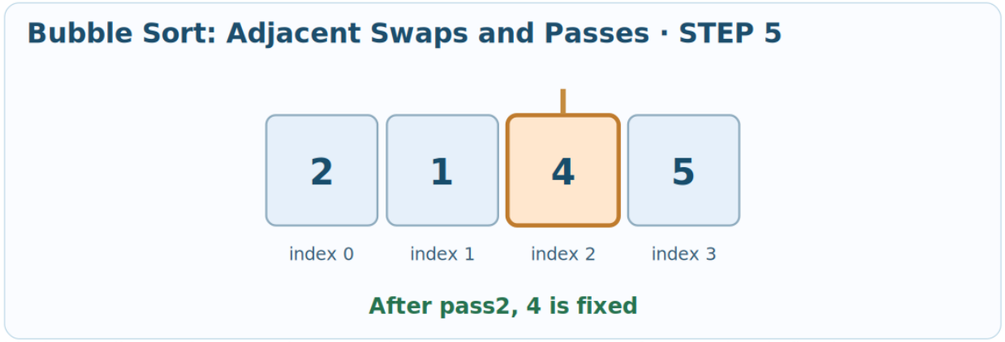


Bubble Sort: Adjacent Swaps and Passes · STEP 4; array 2, 4, 1, 5; After pass1, largest5 fixed

New Learning · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

New Learning



Bubble Sort: Adjacent Swaps and Passes · STEP 5; array 2, 1, 4, 5; After pass2, 4 is fixed

YOU DO — INDEPENDENT PRACTICE

Questions · Part 2

Answer independently. Show code, calculations or a justified explanation.

Question 3

Explain bubble-sort worst-time and extra-space complexity and benefit of early-stop. [6]

YOU DO — INDEPENDENT PRACTICE

Questions · Part 1

Answer independently. Show code, calculations or a justified explanation.

Question 1

Give all states from first pass [5,2,4,1]. [6]

Question 2

Give pass-end arrays for [5,2,4,1] until sorted. [6]

I DO — WORKED EXAMPLE

Follow the worked method

Show full first pass [4,1,3,2].

Check the working

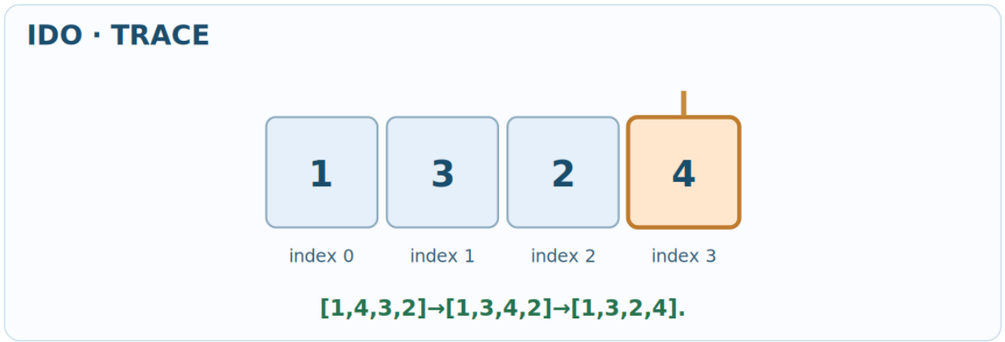
[1,4,3,2]→[1,3,4,2]→[1,3,2,4].

I Do · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

Show full first pass [4,1,3,2].

[1,4,3,2]→[1,3,4,2]→[1,3,2,4].



IDO · TRACE; array 1, 3, 2, 4; [1,4,3,2]→[1,3,4,2]→[1,3,2,4].

WE DO — GUIDED PRACTICE

Try → Discuss → Improve

TRY: Attempt the guided questions in BLACK PEN. DISCUSS: Compare approaches with a partner, explain your reasoning and agree one improvement. IMPROVE: After the Exit Ticket, use the source CHECK model in GREEN PEN to correct and explain errors.

Guided 1 — TRY

Show full first pass [3,2,5,1].
Source code / input:
[2,3,5,1] (swap); 3,5 no swap; [2,3,1,5] (swap).